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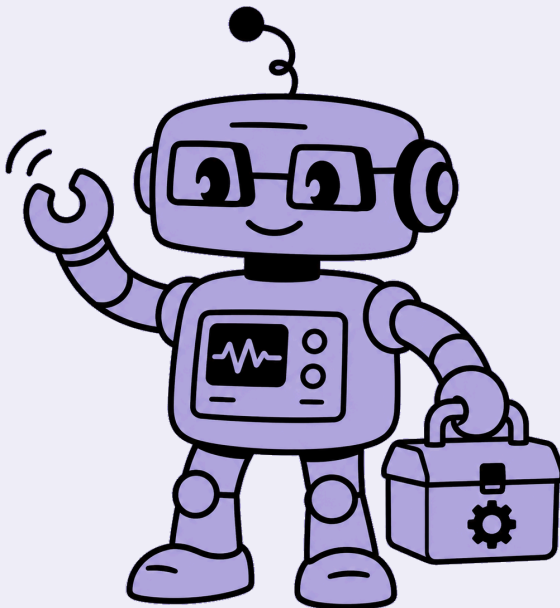
openhouse

robotics • level 2

◆ electronics ◆

your portfolio of the openhouse experience

ages 5-8



why electronics?

This is an **engineering** book. Engineers build real machines and work out how they work.

Inside almost every machine – a torch, a fan, a robot vacuum, an automatic door – is a **circuit**: electricity flowing round a loop of wires.

Electronics is how those circuits work. You'll build machines that light up, move, and even sense the world – and say why they work.

You grow four **skills** and build five real **machines**. You start from the very first circuit – no need to have built machines before. The next pages show the skills, the machines, the words, and how to stay safe.

the skills & their outcomes

the skills you build

You grow all four skills on **every machine**. Each grows in four **outcomes** – the machines climb them a little further each time. **presenting & explaining** is new this level.



observing & understanding

1. notice when it works and when it doesn't →
 2. describe the pattern in words →
 3. compare two circuits →
 4. **predict if it will work, and say why**
-



building & making

1. snap the blocks together from the card →
 2. complete the loop so it works →
 3. build it and name each block's job →
 4. **add or swap a block to change what it does**
-



problem solving

1. notice it doesn't work →
 2. try a fix →
 3. change one thing and see if it helps →
 4. **find the break and get it working**
-



presenting & explaining

1. name your circuit and its blocks →
 2. say what it does →
 3. say why it works →
 4. **show it and answer a question**
-

the machines

the machines you build

Five real machines in one book. Each teaches one big electronics idea – from a first circuit to a machine that senses and responds.

parking barrier

open & closed circuits

wind turbine

polarity

soccer bot

polarity reversal

cleaning bot

one battery, many jobs

sensor-controlled crane

input & output

the words you'll learn

Real engineers use real words. You'll learn the **blocks** (the parts you hold) and the **ideas** (what they do). Tick each one off as you learn it.

the blocks

power block – gives power

jumper wires – carry the electricity

led block – lights up

resistor block – keeps the led safe

switch block – turns it on and off

motor block – makes movement

direction switch – forward or backward

speed dial – changes the speed

sensor block – the robot's eyes

motor driver – like the brain

the ideas

circuit – the complete loop electricity travels round

open & closed – a gap stops it; joined, it works

polarity – electricity has a direction – + and –

series & parallel – share one path, or each its own path

input & output – what a machine senses, and what it does

staying safe with electricity

Electricity is useful, but it must be treated with care. Read these before you start.



The battery blocks in your kit are **safe** to touch and build with.



The power in a **wall socket** is much, much stronger. **Never** poke anything into a socket.



Never touch a switch, plug or wire with **wet hands**.



Wires wear a plastic coat – an **insulator** – to keep the electricity safely inside. Don't use a wire with the coat torn off.



If something smells hot or looks burnt, **stop** and tell an adult.

what's inside this book

For **every machine** you do six things – you can do them on your own:

1

explore

a story, a quick activity, the words, and a fun fact

2

experiment

run the real circuit, then answer a question about it

3

complete the circuit

finish a circuit – add the missing block or wire

4

build & name

build the circuit, then name its blocks

5

solve

fix a circuit that won't work – and stay safe

6

show

show it and say what you learned

At the end – the **level-up build**: a machine that senses and responds, the check you're ready for the next level.



meet the parking barrier

Aarav's building has a car-park gate that lifts when the guard presses a button. One evening it wouldn't open and cars piled up. Aarav spotted a loose wire, showed the guard, and together they fixed the loop – and the gate lifted again.

words to know

circuit**switch****current**

 **your turn:** tick what is a simple circuit:

- ☐ a torch
- ☐ a cushion
- ☐ a doorbell



did you know?

the flow of electricity going round and round the loop is called **current**.



observing & understanding

outcome 1 – you're learning to: notice when it works and when it doesn't



do the experiment

Build a loop: a power block, a switch and an LED. Turn the switch on so it lights. Now take one wire off the loop.

now think: With one wire off, the loop is broken. The light will:



circle your answer:



stay off



still shine



draw what you saw



complete the circuit

a circuit only works when the loop is complete. finish this one:



(the wires join back into a loop)

which block goes in the empty space, to turn the light on and off?



circle the block that completes the circuit:

☐

a switch block

☐

a motor block

☐

a sensor block



building & making

Build your parking barrier circuit from your blocks and circuit card.

Make it so it **lifts and drops when you flip the switch.**

☐

tick when it works

- 1 **Draw** your circuit in the box below.
- 2 **Label** these two blocks on your drawing:

your labels

the power block

the switch block

 draw your parking barrier circuit – then label its blocks



problem solving

outcome 1 – you're learning to: notice it doesn't work

The parking barrier's light won't come on – one wire has come loose.
Find the break and make the loop work again.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

A torch, a doorbell and a table lamp are all simple circuits – a loop the electricity flows around.



presenting & explaining

show your parking barrier to a friend. say three things:

1. **what it is** – “this is my parking barrier. it works as a circuit with a switch.”
2. **its blocks** – point to the power block and the switch block.
3. **what it does** – “it lifts and drops when you flip the switch.”

 the new words i learned – write what each one means:

circuit

switch

current



meet the wind turbine

Meera visited a wind farm on a hill. Her cousin asked how the blades made a light glow. Meera built a little turbine – and when she swapped the wires by mistake, the blades spun backwards. They both laughed, then worked out why.

words to know

motor**led****polarity**

 **your turn:** tick what must go the right way round:

- ☐ a battery in a remote
- ☐ a picture book
- ☐ a torch battery



did you know?

a battery stores **charge** – like a store of energy, waiting to flow.



observing & understanding

outcome 2 – you're learning to: describe the pattern in words



do the experiment

Wire the motor to the power block so it spins. Now swap the two wires over.

now think: When you swap the two wires, the motor spins:



circle your answer:

☐

the other way

☐

the same way



draw what you saw



complete the circuit

a circuit only works when the loop is complete. finish this one:

power block

motor block

led block

(the blocks are apart – no wires yet)

these three blocks are not joined up. draw the wires to make ONE loop, so power flows all the way round.

- ✎ draw the wires between the blocks – then check power can flow all the way round the loop.

✎ re-draw your finished circuit here, with all its wires



building & making

Build your wind turbine circuit from your blocks and circuit card.

Make it so it **spins the blades and lights the led.**

☐

tick when it works

1 Draw your circuit in the box below.

2 Match each block to what it does – draw a line.

the motor block

lights up when power flows the right way

the led block

spins to turn the blades

 now draw your circuit



problem solving

outcome 2 – you're learning to: try a fix

The wind turbine's motor is spinning the wrong way. Change one thing so it spins the way you want.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

Batteries in a remote must go the right way round, or nothing works – that's polarity.



presenting & explaining

show your wind turbine to a friend. say three things:

1. **what it is** – “this is my wind turbine. it shows polarity.”
2. **its blocks** – point to the motor block and the led block.
3. **what it does** – “it spins the blades and lights the led.”

 the new words i learned – write what each one means:

motor

led

polarity



meet the soccer bot

The class built soccer bots for a friendly match. Sam's bot could only drive straight, so it kept missing the ball. A friend showed Sam the direction switch, and soon Sam's bot could turn and join in too.

words to know

direction switch

reverse

steer



your turn: tick what can drive forwards and backwards:

☐

a remote-control car

☐

a brick wall

☐

a toy robot



did you know?

electricity flows from the + side all the way round the loop to the – side.



observing & understanding

outcome 2 – you're learning to: describe the pattern in words



do the experiment

Wire two motors through a direction switch. Drive forward, then flip the switch.

now think: Flipping the direction switch makes the bot:



circle your answer:

☐

reverse

☐

stop for good



draw what you saw



complete the circuit

a circuit only works when the loop is complete. finish this one:



(the wires join back into a loop)

which block lets you flip the motors forwards and backwards?



circle the block that completes the circuit:

☐

a direction switch

☐

an led block

☐

a resistor block



building & making

Build your soccer bot circuit from your blocks and circuit card. Make it so it **drives forwards, back, and turns.**

☐

tick when it works

- 1 **Draw** your circuit in the box below.
- 2 **Fill in each block** – write it from the word bank.

flips the motors forwards or back ...

the _____

drive and steer the bot ...

the _____

word bank

the direction switch

the two motors

 now draw your circuit



problem solving

outcome 2 – you're learning to: try a fix

The soccer bot only drives straight – it won't turn. Change one thing so it can steer.



Draw your fix – what did you change to make it work?



draw your fix here



in real life

A remote-control car changes direction the same way – a switch flips the motor.



presenting & explaining

show your soccer bot to a friend. say three things:

1. **what it is** – “this is my soccer bot. it reverses its motors to steer.”
2. **its blocks** – point to the direction switch and the two motors.
3. **what it does** – “it drives forwards, back, and turns.”

 the new words i learned – write what each one means:

direction switch _____

reverse _____

steer _____



meet the cleaning bot

Priya's grandmother finds sweeping tiring. Priya built a cleaning bot that runs its wheels and brush together from one battery, to help her grandmother keep the floor clean.

words to know

series

parallel

speed



your turn: tick what runs many things from one battery:

☐

a robot vacuum

☐

a garden rock

☐

a mobile phone



did you know?

voltage is the push behind the flow – more push makes the motors spin faster.



observing & understanding

outcome 3 – you're learning to: compare two circuits



do the experiment

Run two motors in one path (series), then give each its own path (parallel).
Watch their speed.

now think: Two motors sharing ONE path (series) each run:



circle your answer:



a bit slower



much faster



draw what you saw



complete the circuit

a circuit only works when the loop is complete. finish this one:

battery

motor

motor

(the blocks are apart – no wires yet)

draw the wires so BOTH motors get their OWN path from the battery
– that makes a parallel circuit.



draw the wires between the blocks – then check power can flow all the way round the loop.



re-draw your finished circuit here, with all its wires



building & making

Build your cleaning bot circuit from your blocks and circuit card.

Make it so it **runs all its motors from one battery**.

☐

tick when it works

- 1 **Draw** your circuit in the box below.
- 2 **Label** these two blocks on your drawing:

your labels

the battery

the speed dial

 draw your cleaning bot circuit – then label its blocks



problem solving

outcome 3 – you're learning to: change one thing and see if it helps

One of the cleaning bot's motors won't run; the others are fine. Find out why and fix it.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

A robot vacuum runs its wheels and brush from one battery, just like your cleaning bot.



presenting & explaining

show your cleaning bot to a friend. say three things:

1. **what it is** – “this is my cleaning bot. it runs many motors from one battery.”
2. **its blocks** – point to the battery and the speed dial.
3. **what it does** – “it runs all its motors from one battery.”

 the new words i learned – write what each one means:

series

parallel

speed



meet the sensor-controlled crane

At the port, huge cranes stop on their own if someone walks underneath. Reza built a small sensor crane that stops when a hand comes near – a safe crane, he told his little sister, so no one gets hurt.

words to know

sensor

input

output



your turn: tick what senses and responds on its own:

☐

an automatic tap

☐

a wooden chair

☐

a night-light



did you know?

a **sensor** turns what it sees into a tiny electric signal the driver can read.



observing & understanding

outcome 4 – you're learning to: predict if it will work, and say why



do the experiment

Wire an IR sensor to the motor driver to the motor. Wave your hand in front of the sensor.

now think: The sensor sees your hand and tells the driver to:



circle your answer:

☐

change the motor

☐

do nothing

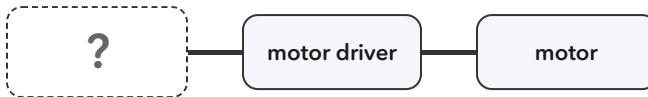


draw what you saw



complete the circuit

a circuit only works when the loop is complete. finish this one:



(the wires join back into a loop)

which block is the INPUT – the part that senses and starts it off?



circle the block that completes the circuit:

- ☐ a sensor block
- ☐ an led block
- ☐ a resistor block



building & making

Build your sensor-controlled crane circuit from your blocks and circuit card. Make it so it **stops on its own when your hand is near**.

☐

tick when it works

1 Draw your circuit in the box below.

2 Match each block to what it does – draw a line.

the sensor

the brain – it switches the motor

the motor driver

the eyes – it senses your hand

 now draw your circuit



problem solving

outcome 4 – you're learning to: find the break and get it working

The crane doesn't stop when your hand is near. Change one thing so the sensor makes it stop.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

Automatic taps, doors and night-lights all sense and respond – input to output.



presenting & explaining

show your sensor-controlled crane to a friend. say three things:

1. **what it is** – “this is my sensor-controlled crane. it senses and responds.”
2. **its blocks** – point to the sensor and the motor driver.
3. **what it does** – “it stops on its own when your hand is near.”

 the new words i learned – write what each one means:

sensor

input

output



your level-up build

This is your **level check**. Build **one machine that senses and responds** – a circuit that reacts to a sensor, or your own circuit that does a real job. Show all four skills on it.

✓ **you are ready for the next level when you can:**

- ☐ predict if it will work, and say why
- ☐ add or swap a block to change what it does
- ☐ find the break and get it working
- ☐ show it and answer a question

draw your level-up build here